Kinematic and Singularity Analysis Report with Explicit Matrix Operations

Automated Generation System

March 19, 2026

1 Mathematical Formulation & Singularity Extraction

The closed-loop spatial mechanism is driven by q_1, leaving $q_2 \dots q_7$ as 6 passive joints. To evaluate the structural singularity of the entire mechanism loop, we extract the passive Jacobian $J_p \in R^{6\times 6}$ corresponding to the remaining kinematics loop.

By identifying the local minima of the 6th singular value (σ_6), the following critical singular configurations (dead centers) were numerically extracted:

- Driven by $q_1 = 210.58°$, the minimum singular value drops to $\sigma_6 = 0.001481$.

2 Example Matrix Calculation at $q_1 = 0°$

The solved full joint configuration (in radians) is:

$$\mathbf{q} = \begin{bmatrix} 0.0000 \\ 0.0276 \\ -0.0003 \\ -0.0552 \\ 0.0005 \\ -0.0276 \\ -0.0008 \end{bmatrix} \tag{1}$$

The full 6×7 Jacobian matrix J:

$$J = \begin{bmatrix} 0.000 & 0.000 & 0.000 & 2.499 & -0.069 & -2.501 & -0.000 \\ 0.000 & -1.000 & -0.000 & 2.999 & -0.000 & 1.997 & 0.000 \\ -0.000 & 0.000 & -3.000 & 0.069 & -0.500 & 0.069 & -2.000 \\ 0.000 & 0.000 & 0.000 & 0.028 & 0.000 & 0.028 & -0.001 \\ 0.000 & 0.000 & -1.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 1.000 & 0.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{2}$$

The structural 6×6 passive Jacobian J_p (excluding J_1):

$$J_p = \begin{bmatrix} 0.000 & 0.000 & 2.499 & -0.069 & -2.501 & -0.000 \\ -1.000 & -0.000 & 2.999 & -0.000 & 1.997 & 0.000 \\ 0.000 & -3.000 & 0.069 & -0.500 & 0.069 & -2.000 \\ 0.000 & 0.000 & 0.028 & 0.000 & 0.028 & -0.001 \\ 0.000 & -1.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 0.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{3}$$

Applying SVD on $J_p = U\Sigma V^T$ yields 6 singular values. The left singular vectors U:

$$U = \begin{bmatrix} 0.665 & 0.106 & -0.635 & -0.002 & 0.379 & 0.002 \\ 0.671 & -0.031 & 0.738 & 0.007 & 0.068 & 0.011 \\ 0.064 & -0.983 & -0.097 & -0.142 & -0.002 & 0.000 \\ 0.005 & -0.001 & 0.009 & -0.000 & 0.012 & -1.000 \\ 0.010 & -0.140 & -0.023 & 0.990 & -0.011 & -0.000 \\ -0.323 & -0.045 & 0.206 & 0.012 & 0.923 & 0.011 \end{bmatrix} \tag{4}$$

The singular values matrix Σ (6×6):

$$\Sigma = \begin{bmatrix} 4.178 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 3.672 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 3.351 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.669 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.892 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \end{bmatrix} \tag{5}$$

And V^T (6×6):

$$V^T = \begin{bmatrix} -0.238 & -0.048 & 0.957 & -0.021 & -0.153 & -0.028 \\ -0.004 & 0.841 & 0.041 & 0.170 & -0.119 & 0.497 \\ -0.159 & 0.094 & 0.123 & 0.035 & 0.973 & 0.051 \\ 0.003 & -0.338 & -0.004 & -0.550 & 0.013 & 0.763 \\ 0.958 & 0.018 & 0.258 & -0.016 & 0.123 & -0.008 \\ -0.023 & 0.408 & -0.014 & -0.816 & 0.009 & -0.408 \end{bmatrix} \tag{6}$$

3 Analysis Results & Curve Fitting

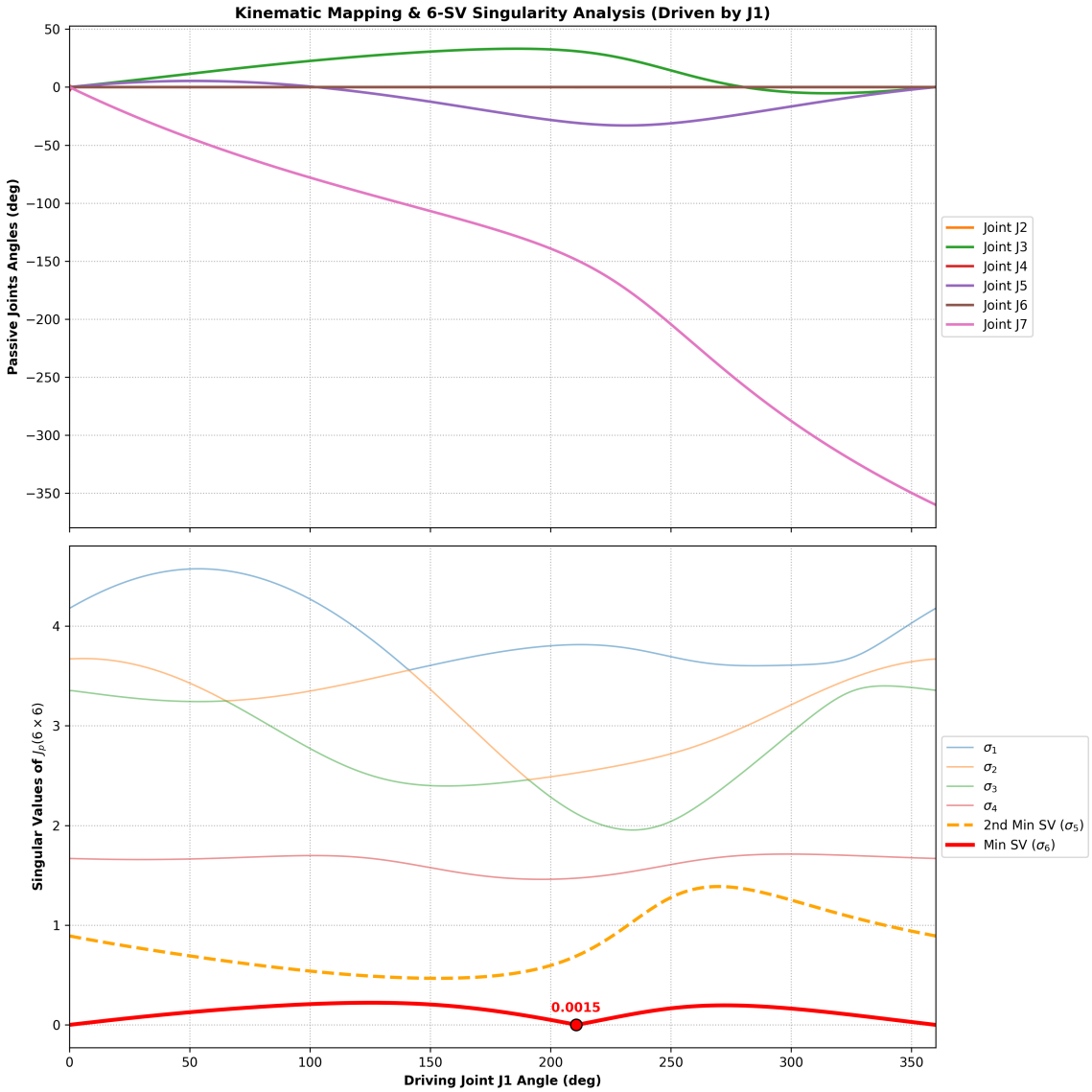


Figure 1: Kinematic mapping and full 6×6 singular value evolution. Critical dead centers are highlighted in red.

The 3rd-order Fourier series expressions for the passive joints are:

- **Joint J2:**
  $Q_{J2}(x) = +0.004 + 0.008\cos(x) - 0.000\sin(x) + 0.008\cos(2x) + 0.001\sin(2x) + 0.008\cos(3x) - 0.001\sin(3x)$
- **Joint J3:**
  $Q_{J3}(x) = +14.740 - 17.160\cos(x) + 7.200\sin(x) + 2.093\cos(2x) + 3.424\sin(2x) + 0.617\cos(3x) - 0.949\sin(3x)$
- **Joint J4:**
  $Q_{J4}(x) = -0.009 - 0.015\cos(x) + 0.001\sin(x) - 0.017\cos(2x) - 0.001\sin(2x) - 0.016\cos(3x) + 0.001\sin(3x)$

- **Joint J5:**
  $Q_{J5}(x) = -11.728 + 11.634\cos(x) + 14.541\sin(x) + 0.433\cos(2x) - 1.922\sin(2x) - 0.459\cos(3x) + 0.224\sin(3x)$

- **Joint J6:**
  $Q_{J6}(x) = -0.004 - 0.008\cos(x) + 0.000\sin(x) - 0.008\cos(2x) - 0.001\sin(2x) - 0.008\cos(3x) + 0.001\sin(3x)$

- **Joint J7:**
  $Q_{J7}(x) = -153.532 - 28.794\cos(x) + 107.247\sin(x) + 1.660\cos(2x) + 62.635\sin(2x) + 1.077\cos(3x) + 37.015\sin(3x)$